

Lab 01: linear algebra + computing review

Ikea furniture

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2026-08-31

Setup

Load packages and data:

```
library(tidyverse)
library(tidymodels)
ikea <- read_csv("data/ikea.csv")
```

Linear algebra review

[Delete the placeholders for Exercises 1 - 6 if you are submitting handwritten responses for these exercises]

Exercise 1

Attached at end of pdf

Exercise 2

Attached at end of pdf

Exercise 3

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Exercise 4

Attached at end of pdf

Exercise 5

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Exercise 6

Attached at end of pdf

Computing review

Exercise 7

```
library(tidyverse)
library(knitr)
ikea <- read_csv("data/ikea.csv")
```

Rows: 3694 Columns: 13

-- Column specification -----

Delimiter: ","

chr (6): name, category, link, other_colors, short_description, designer

dbl (6): X1, item_id, depth, height, width, price_usd

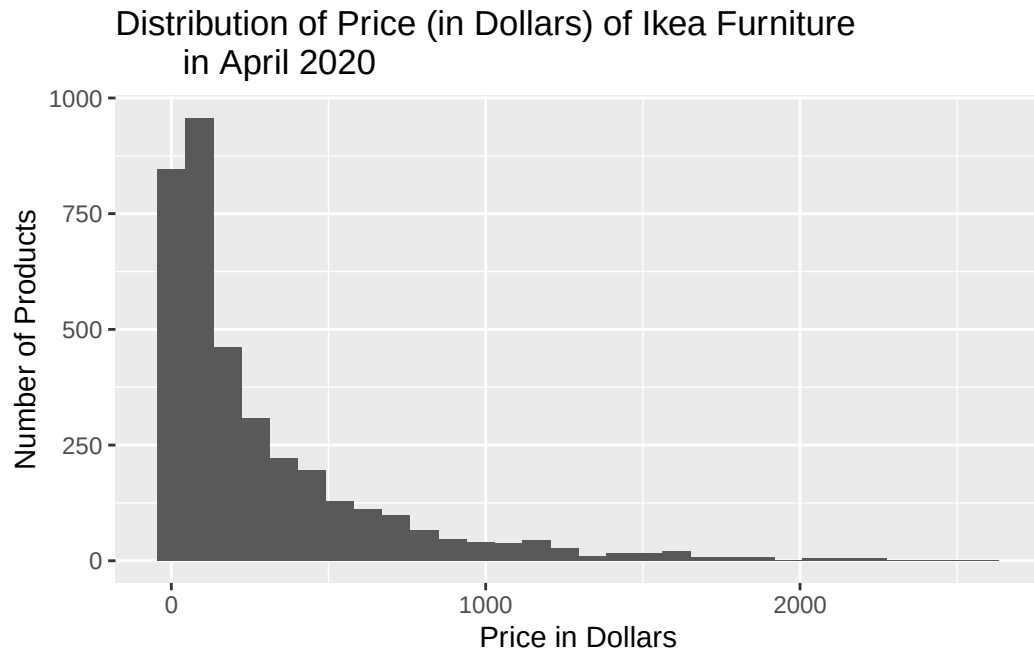
lgl (1): sellable_online

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
ggplot(data = ikea, aes(x = price_usd)) +
  geom_histogram() +
  labs(x = "Price in Dollars",
       y = "Number of Products",
       title = "Distribution of Price (in Dollars) of Ikea Furniture
               in April 2020")
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Exercise 8

The median is more representative of the center than the mean as the distribution is right skewed with a few outliers with really high cost. This means that the mean would be “stretched” higher due to those really high cost items, when in reality there are only a few of them. For median, those outliers have significantly less impact, therefore the median is a better representative of the center of the distribution.

Exercise 9

```
ikea_sub <- ikea |>
  filter(category %in% c( "Sofas & armchairs",
                        "Bookcases & shelving units"))

ikea_sub <- ikea_sub |>
  drop_na(width, price_usd)
```

There are 765 observations with 13 variables. The advantage of deleting all observations with one or more missing variables is that we don't have to deal with NA values during computing of data. A disadvantage would be that we have a smaller sample size as we are completely removing the row, which could lead to things such as decreasing power.

Exercise 10

```
ggplot(data = ikea_sub, aes(x = width, y = price_usd)) +  
  geom_point() +  
  labs(x = "Width of Furniture (in cm)",  
       y = "Price in USD",  
       title = "Relationship Between Price (in USD) and Width (in cm) of Ikea Furniture in April 2020")
```

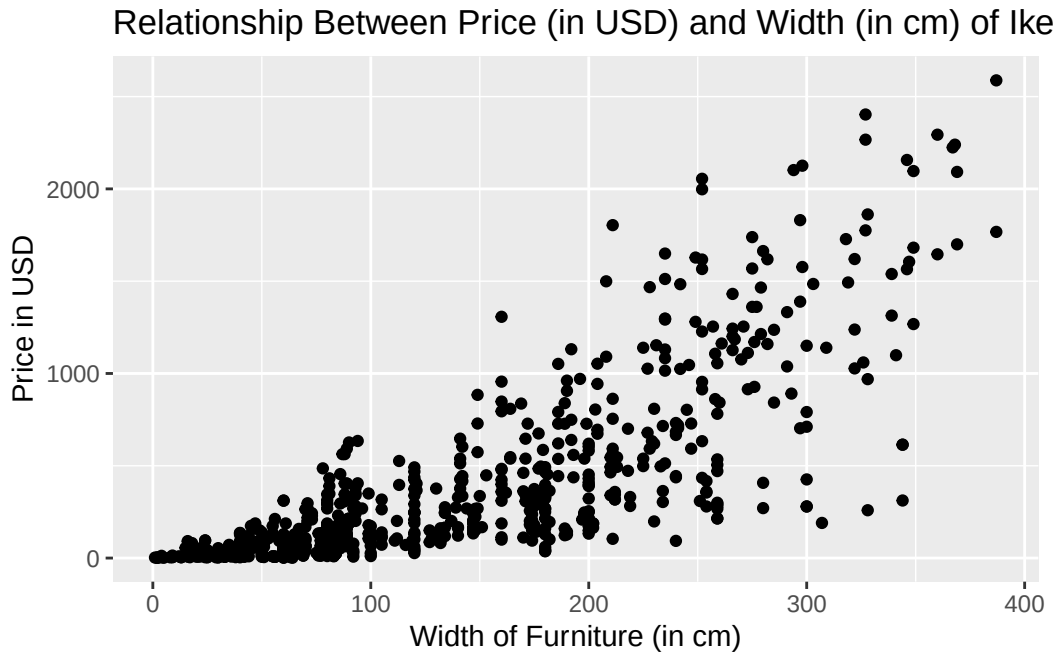


Figure 1: There is a positive nonlinear relationship between Price and Width of Ikea Furniture

```
ggplot(data = ikea_sub, aes(x = width, y = price_usd, shape = category, color = category)) +  
  geom_point() +  
  labs(x = "Width of Furniture (in cm)",  
       y = "Price in USD",  
       title = "Relationship Between Price (in USD) and Width (in cm) of Ikea  
Furniture in April 2020",  
       color = "Furniture Category",  
       shape = "Furniture Category")
```

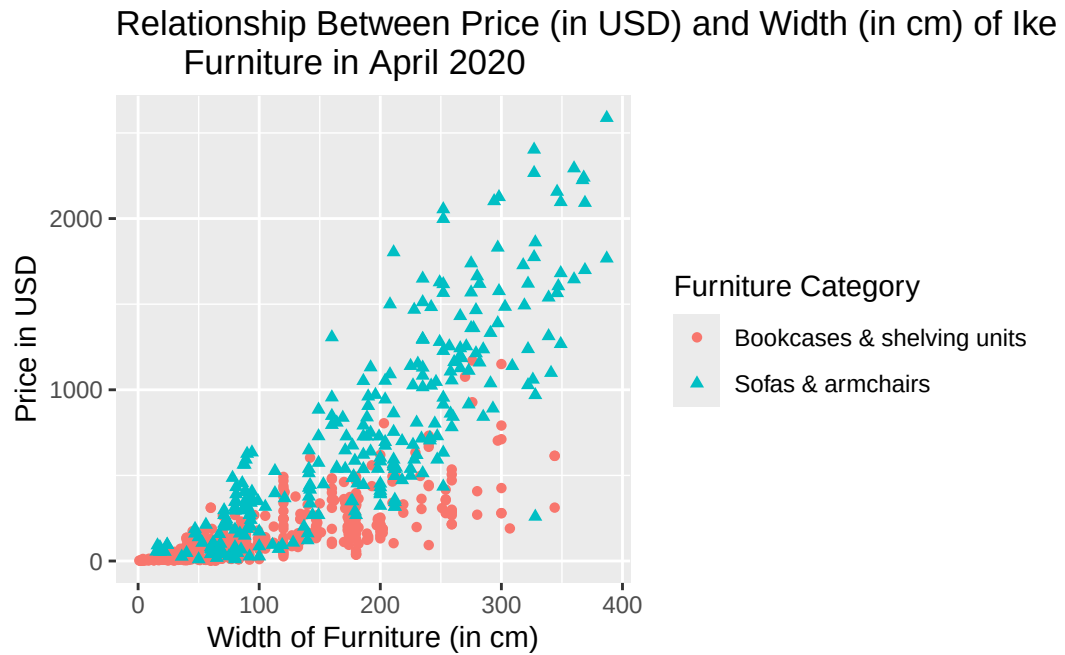


Figure 2: As the width increases, the price of sofas and armchairs increase at a faster rate compared to bookcases and shelving units

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